

5.3.9.2 PSI oscillator characteristics

Table 36. PSI oscillator characteristics

Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max	Unit
f _{PSI}	PSI potential frequency	If reference is HSE	-	100	-	MHz
		at 8, 16, 24, 32 or 48 MHz	-	144	-	
		or HSDIV18	-	160 ⁽²⁾	-	
		If reference is HSE at 25 or 50 MHz	-	100	-	
			-	141.67	-	
			-	158.33 ⁽²⁾	-	
		If reference is LSE at 32 KHz	-	100.008	-	
			-	144.015	-	
			-	160.006 ⁽²⁾	-	
DuCy(PSI) ⁽³⁾	Duty Cycle	-	45		55	%
t _{su} (PSI) ⁽⁴⁾	PSI startup time	On 32 KHz	-	850	1750	μs
	PSI startup time	On 8 MHz	-	25	45	μs
I _{DD} (PSI) ⁽³⁾⁽⁵⁾	PSI supply regulation bloc oscillator power consumption	If HSI not ON	-	91	-	μA
	PSI oscillator power consumption	100 MHz	-	95	-	
	PSI oscillator power consumption	144 MHz	-	68	-	
	PSI oscillator power consumption	160 MHz	-	77	-	
N _T jitter ⁽³⁾	Next transition jitter ⁽⁶⁾	On PSIDIV3 (48 MHz with 32 kHz CK_IN)	-	46.8	264	ps
		On PSIDIV3 (48 MHz with 8 MHz CK_IN)	-	52.6	276	ps
P _T jitter ⁽³⁾	Paired transition jitter ⁽⁷⁾	On PSIDIV3 (48 MHz with 32 kHz CK_IN)	-	54.4	331	ps
		On PSIDIV3 (48 MHz with 8 MHz CK_IN)	-	60.6	367	ps
P _{er} jitter ⁽³⁾	Period Jitter standard deviation	On PSIS (100 MHz)	-	15.5	-	ps
		On PSIDIV3 (33.33 MHz)	-	27	-	ps
		On PSIS (144 MHz)	-	14	-	ps
		On PSIDIV3 (48 MHz)	-	24.5	-	ps
		On PSIS (160 MHz)	-	13.5	-	ps
		On PSIDIV3 (53.33 MHz)	-	23	-	ps
LT jitter ⁽³⁾	Long term jitter ethernet	On PSIS (100 MHz with 32 kHz CK_IN)	-	-	13.7 (RMS) 96.7 (peak)	ns
		On PSIS	-	-	0.79 (RMS)	ns